AQA A-Level Further Mathematics (7367) — Topic 1.9

1.9: Differential Equations

1. First Order Separable Differential Equations

A differential equation is **separable** if it can be written as:

$$\frac{dy}{dx} = f(x)\,g(y)$$

Rearranging: $\displaystyle\int \frac{1}{g(y)}\,dy = \int f(x)\,dx$

Example: Solve $\dfrac{dy}{dx} = xy$, with $y(0) = 2$.

$\int \frac{1}{y}\,dy = \int x\,dx \Rightarrow \ln|y| = \frac{x^2}{2} + C$
$|y| = Ae^{x^2/2}$, so $y = Ae^{x^2/2}$ (where A can be positive or negative).

Applying $y(0) = 2$: $2 = A$. Particular solution: $y = 2e^{x^2/2}$.

Example: Solve $\dfrac{dy}{dx} = \dfrac{1+y^2}{1+x^2}$.

$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$
$\arctan y = \arctan x + C$
$y = \tan(\arctan x + C)$

2. First Order Linear Differential Equations

Standard form: $\dfrac{dy}{dx} + P(x)\,y = Q(x)$

Integrating factor: $\mu = e^{\int P(x)\,dx}$

Multiply through by μ: the left side becomes $\dfrac{d}{dx}(\mu y)$, giving:

$$\mu y = \int \mu\, Q(x)\,dx$$

Example: Solve $\dfrac{dy}{dx} + \dfrac{2y}{x} = x^2$.

$P(x) = 2/x$. Integrating factor: $\mu = e^{\int 2/x\,dx} = e^{2\ln x} = x^2$.

Multiply: $\dfrac{d}{dx}(x^2 y) = x^4$
Integrate: $x^2 y = \dfrac{x^5}{5} + C$
General solution: $y = \dfrac{x^3}{5} + \dfrac{C}{x^2}$

AQA Exam Tip: Always simplify the integrating factor $\mu = e^{\int P\,dx}$ before multiplying through. A common error is forgetting to multiply $Q(x)$ on the right-hand side.

3. Second Order Homogeneous Differential Equations

Form: $a\dfrac{d^2y}{dx^2} + b\dfrac{dy}{dx} + cy = 0$

Auxiliary equation: $am^2 + bm + c = 0$

The general solution depends on the roots of the auxiliary equation:

Roots of auxiliary equation	General solution
Two distinct real roots m_1, m_2	$y = Ae^{m_1 x} + Be^{m_2 x}$
Repeated real root m	$y = (A + Bx)e^{mx}$
Complex roots $\alpha \pm \beta i$	$y = e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

Example (distinct real roots): Solve $\dfrac{d^2y}{dx^2} - 5\dfrac{dy}{dx} + 6y = 0$.

Auxiliary: $m^2 - 5m + 6 = 0 \Rightarrow (m-2)(m-3) = 0 \Rightarrow m = 2, 3$
General solution: $y = Ae^{2x} + Be^{3x}$

Example (repeated root): Solve $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 9y = 0$.

Auxiliary: $m^2 - 6m + 9 = (m-3)^2 = 0 \Rightarrow m = 3$ (repeated)
General solution: $y = (A + Bx)e^{3x}$

Example (complex roots): Solve $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 13y = 0$.

Auxiliary: $m^2 - 4m + 13 = 0 \Rightarrow m = \dfrac{4 \pm \sqrt{16-52}}{2} = 2 \pm 3i$
So $\alpha = 2$, $\beta = 3$.
General solution: $y = e^{2x}(A\cos 3x + B\sin 3x)$

4. Second Order Non-Homogeneous Differential Equations

Form: $ay'' + by' + cy = f(x)$

General solution: $y = y_{CF} + y_{PI}$
where y_{CF} is the complementary function (solution of the homogeneous equation) and y_{PI} is any particular integral satisfying the full equation.

Choosing a Particular Integral

$f(x)$	Trial y_{PI}
ke^{px} (where e^{px} is not in CF)	λe^{px}
ke^{px} (where e^{px} is in CF — simple root)	$\lambda x e^{px}$
ke^{px} (where e^{px} is in CF — repeated root)	$\lambda x^2 e^{px}$
$k\sin(px)$ or $k\cos(px)$	$\lambda\cos(px) + \mu\sin(px)$
Polynomial of degree n	Polynomial of degree n: $a_n x^n + \cdots + a_0$
Product: polynomial × exponential	Same product form with unknown coefficients

Example: Find the general solution of $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = 4e^{3x}$.

CF: Auxiliary: $m^2 - 3m + 2 = (m-1)(m-2) = 0$, so $y_{CF} = Ae^x + Be^{2x}$.

PI: Try $y_{PI} = \lambda e^{3x}$.
$y' = 3\lambda e^{3x}$, $y'' = 9\lambda e^{3x}$
Substitute: $9\lambda e^{3x} - 9\lambda e^{3x} + 2\lambda e^{3x} = 4e^{3x}$
$2\lambda = 4 \Rightarrow \lambda = 2$

General solution: $y = Ae^x + Be^{2x} + 2e^{3x}$

Example: Find the general solution of $y'' - 3y' + 2y = \sin x$.

CF: $y_{CF} = Ae^x + Be^{2x}$ (as before).

PI: Try $y_{PI} = \lambda \cos x + \mu \sin x$.
$y' = -\lambda \sin x + \mu \cos x$, $y'' = -\lambda \cos x - \mu \sin x$
Substitute:
$(-\lambda \cos x - \mu \sin x) - 3(-\lambda \sin x + \mu \cos x) + 2(\lambda \cos x + \mu \sin x) = \sin x$
Cosine: $-\lambda - 3\mu + 2\lambda = \lambda - 3\mu = 0$
Sine: $-\mu + 3\lambda + 2\mu = \mu + 3\lambda = 1$
From first: $\lambda = 3\mu$. Substitute: $\mu + 9\mu = 1 \Rightarrow \mu = \frac{1}{10}$, $\lambda = \frac{3}{10}$.

$$y = Ae^x + Be^{2x} + \frac{3}{10}\cos x + \frac{1}{10}\sin x$$

AQA Exam Tip: If your trial PI is part of the complementary function, you must multiply the trial by x (or x^2 for repeated roots). Forgetting this gives a zero coefficient and no PI.

5. Applying Initial/Boundary Conditions

Example: Solve $y'' + 4y = 8$ with $y(0) = 3$, $y'(0) = 0$.

CF: Auxiliary: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$. $y_{CF} = A\cos 2x + B\sin 2x$

PI: Try $y_{PI} = \lambda$ (constant). Then $y'' = 0$: $0 + 4\lambda = 8 \Rightarrow \lambda = 2$.

General solution: $y = A\cos 2x + B\sin 2x + 2$

Apply $y(0) = 3$: $A + 2 = 3 \Rightarrow A = 1$
$y' = -2A\sin 2x + 2B\cos 2x$; $y'(0) = 0$: $2B = 0 \Rightarrow B = 0$

Particular solution: $y = \cos 2x + 2$

6. Coupled Differential Equations

A system of coupled first-order equations can often be reduced to a single second-order equation.

Method for coupled ODEs: Given $\frac{dx}{dt} = ax + by$ and $\frac{dy}{dt} = cx + dy$:

1. Differentiate one equation to get $\ddot{x}$
2. Substitute the other equation to eliminate y (and $\dot{y}$)
3. Solve the resulting second-order ODE for $x(t)$
4. Back-substitute to find $y(t)$

Example: Solve $\dot{x} = 3x + 2y$, $\dot{y} = x + 2y$, with $x(0) = 1$, $y(0) = 0$.

Differentiate first equation: $\ddot{x} = 3\dot{x} + 2\dot{y} = 3\dot{x} + 2(x + 2y)$
From first equation: $y = \frac{\dot{x} - 3x}{2}$
Substitute: $\ddot{x} = 3\dot{x} + 2x + 2 \cdot 2 \cdot \frac{\dot{x} - 3x}{2} = 3\dot{x} + 2x + 2\dot{x} - 6x = 5\dot{x} - 4x$
$\ddot{x} - 5\dot{x} + 4x = 0$
Auxiliary: $m^2 - 5m + 4 = (m-1)(m-4) = 0$
$x = Ae^t + Be^{4t}$
$y = \frac{\dot{x} - 3x}{2} = \frac{Ae^t + 4Be^{4t} - 3Ae^t - 3Be^{4t}}{2} = -Ae^t + \frac{Be^{4t}}{2}$

Applying IC: $A + B = 1$, $-A + B/2 = 0 \Rightarrow B = 2A$, so $3A = 1 \Rightarrow A = 1/3$, $B = 2/3$.
$x = \frac{1}{3}e^t + \frac{2}{3}e^{4t}$, $y = -\frac{1}{3}e^t + \frac{1}{3}e^{4t}$

7. Equilibrium Points and Stability

For a first-order autonomous DE $\frac{dy}{dx} = f(y)$, **equilibrium points** (or fixed points) satisfy $f(y^*) = 0$.

- **Stable equilibrium:** If $f'(y^*) < 0$, small perturbations decay back to y^*
- **Unstable equilibrium:** If $f'(y^*) > 0$, perturbations grow away from y^*

8. Summary of Key Methods

Type	Method
Separable 1st order	Separate variables; integrate both sides
Linear 1st order	Integrating factor $\mu = e^{\int P\,dx}$
2nd order homogeneous	Auxiliary equation; CF depends on roots
2nd order non-homogeneous	General solution = CF + PI; choose PI from table
Coupled 1st order	Differentiate and eliminate to get 2nd order
Particular solution	Apply initial/boundary conditions to general solution

9. Practice Checklist

- Can you solve separable first-order ODEs?
- Can you find and apply an integrating factor?
- Can you solve second-order homogeneous ODEs for all three root types (distinct, repeated, complex)?
- Can you select the correct trial function for a particular integral and find its coefficients?
- Can you deal with the case where the trial PI is part of the CF?
- Can you find a particular solution given initial conditions?
- Can you reduce coupled first-order ODEs to a single second-order ODE?

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Particular Integrals — Complete Table

For $ay'' + by' + cy = f(x)$, choose a trial particular integral based on $f(x)$:

$f(x)$	Trial PI (standard case)	Trial PI (resonance: trial is in CF)
$ke^{\alpha x}$	$\lambda e^{\alpha x}$	$\lambda x e^{\alpha x}$
$k\cos\omega x$ or $k\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$	$x(\lambda\cos\omega x + \mu\sin\omega x)$
kx^n (polynomial deg n)	$\lambda_n x^n + \cdots + \lambda_0$ (full deg n)	$x(\lambda_n x^n + \cdots + \lambda_0)$
$ke^{\alpha x}\cos\omega x$	$e^{\alpha x}(\lambda\cos\omega x + \mu\sin\omega x)$	$xe^{\alpha x}(\lambda\cos\omega x + \mu\sin\omega x)$
$ke^{\alpha x}\cdot x^n$	$e^{\alpha x}(\lambda_n x^n + \cdots + \lambda_0)$	$xe^{\alpha x}(\lambda_n x^n + \cdots + \lambda_0)$

Resonance rule: If the trial PI for $f(x) = ke^{\alpha x}$ (or $\sin$ / $\cos$) is already a term in the CF, multiply the trial by x. If it is still in the CF, multiply by x^2.

Example — resonance: Solve $y'' + 4y = 8\sin 2x$. CF: $y_c = A\cos 2x + B\sin 2x$ (roots $\pm 2i$). Since $\sin 2x$ is in CF, resonance applies. Trial PI: $y_p = x(a\cos 2x + b\sin 2x)$.

$$y_p' = (a + 2bx)\cos 2x + (b - 2ax)\sin 2x$$

$$y_p'' = (4b - 4ax)\cos 2x + (-4a - 4bx)\sin 2x$$

$$y_p'' + 4y_p = 4b\cos 2x - 4a\sin 2x = 8\sin 2x$$

So $a = -2$, $b = 0$. PI: $y_p = -2x\cos 2x$. General solution: $y = A\cos 2x + B\sin 2x - 2x\cos 2x$.

Coupled Differential Equations — Worked Example

To solve a pair of coupled first-order ODEs, use the **elimination method**:

1. Express one variable in terms of the other and its derivative from the first equation.
2. Differentiate and substitute into the second equation to get a single second-order ODE.

3. Solve the second-order ODE, then back-substitute for the second variable.

Example: Solve $\dot{x} = 3x - y$, $\dot{y} = x + y$. Step 1: From the first equation, $y = 3x - \dot{x}$. Step 2: Differentiate: $\dot{y} = 3\dot{x} - \ddot{x}$. Substitute into second: $3\dot{x} - \ddot{x} = x + (3x - \dot{x}) = 4x - \dot{x}$.

$$\ddot{x} - 4\dot{x} + 4x = 0$$

Step 3: Auxiliary equation: $(m-2)^2 = 0$, so $m = 2$ (repeated root).

$$x = (At + B)e^{2t}$$

Step 4: $y = 3x - \dot{x}$. Compute $\dot{x} = (A + 2At + 2B)e^{2t}$. Then:

$$y = 3(At + B)e^{2t} - (A + 2At + 2B)e^{2t} = (At - A + B)e^{2t}$$

Check: Always verify by substituting back into *both* original equations.

Second Order ODE — Physical Interpretation

The equation $m\ddot{x} + c\dot{x} + kx = F(t)$ models a mass-spring-damper system:

- m: mass, c: damping coefficient, k: spring constant, $F(t)$: forcing function.
- Dividing by m: $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = F(t)/m$ where $\omega_0 = \sqrt{k/m}$ is the natural frequency and $\beta = c/(2m)$ is the damping ratio.

Condition	Root type	Motion type
$c^2 < 4mk$ (i.e.\ $\beta < \omega_0$)	Complex conjugates $-\beta \pm i\omega_d$	**Underdamped**: damped oscillation $e^{-\beta t}\sin(\omega_d t + \phi)$
$c^2 = 4mk$	Repeated real root $-\beta$	**Critically damped**: fastest decay without oscillation $(A + Bt)e^{-\beta t}$
$c^2 > 4mk$	Two distinct real negative roots	**Overdamped**: slow exponential decay $Ae^{-\alpha_1 t} + Be^{-\alpha_2 t}$

$c = 0$	Pure imaginary $\pm i\omega_0$	**Simple harmonic motion**: $A\cos\omega_0 t + B\sin\omega_0 t$

Resonance: When the forcing frequency equals the natural frequency ($F(t) = F_0\cos\omega_0 t$, $c = 0$), the response grows without bound: $x_p = \dfrac{F_0 t}{2m\omega_0}\sin\omega_0 t$. This is why resonance is physically dangerous.

AQA Exam Technique — Differential Equations

Key tips:

- **Second-order CF:** Always factorise the auxiliary equation. Three cases: two distinct real roots; repeated root; complex roots $\alpha \pm \beta i$ giving $e^{\alpha x}(A\cos\beta x + B\sin\beta x)$.
- **PI selection:** Check whether the trial PI is already in the CF before choosing. If it is, multiply by x.
- **Initial conditions:** Apply to the *general solution* (CF + PI), not just the CF. Differentiate y_{GS} for the second condition.
- **Coupled equations:** After elimination, solve the resulting ODE first, then back-substitute for the other variable — do not guess its form independently.
- **Integrating factor:** For $y' + P(x)y = Q(x)$, the IF is $e^{\int P\,dx}$. Multiply both sides and recognise the left-hand side as $\frac{d}{dx}[y \cdot \text{IF}]$.

Type	CF	Typical PI form
$y'' + by' + cy = ke^{\alpha x}$	Depends on roots	$\lambda e^{\alpha x}$ (or $\lambda x e^{\alpha x}$ if resonant)
$y'' + by' + cy = k\sin\omega x$	Depends on roots	$\lambda\cos\omega x + \mu\sin\omega x$ (or $x\cdot$ if resonant)
$y'' + by' + cy = kx^n$	Depends on roots	Full polynomial of degree n
Euler equation $x^2y'' + xy' + ny = 0$	Via $x = e^t$	Reduce to constant-coefficient

First-Order ODEs — Full Methods

Method 1 — Separable equations: $\dfrac{dy}{dx} = f(x)g(y)$.

Separate: $\dfrac{1}{g(y)}\,dy = f(x)\,dx$. Integrate both sides.

Example: $\dfrac{dy}{dx} = xy$. Separate: $\dfrac{1}{y}\,dy = x\,dx$.

$\ln|y| = \dfrac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}$ where $A = \pm e^C$.

Method 2 — Integrating factor (linear first-order): $\dfrac{dy}{dx} + P(x)y = Q(x)$.

Multiply by integrating factor $\mu = e^{\int P\,dx}$:

$\dfrac{d}{dx}(\mu y) = \mu Q \Rightarrow \mu y = \int \mu Q\,dx$.

Example: $\dfrac{dy}{dx} + 2y = e^{-x}$.

$\mu = e^{\int 2\,dx} = e^{2x}$. Multiply: $\dfrac{d}{dx}(e^{2x}y) = e^{2x} \cdot e^{-x} = e^x$.

Integrate: $e^{2x}y = e^x + C \Rightarrow y = e^{-x} + Ce^{-2x}$.

Example 2 (non-constant P): $x\dfrac{dy}{dx} + y = x^2$, i.e. $\dfrac{dy}{dx} + \dfrac{1}{x}y = x$.

$\mu = e^{\int (1/x)dx} = e^{\ln x} = x$. Multiply: $\dfrac{d}{dx}(xy) = x^2$.

Integrate: $xy = \dfrac{x^3}{3} + C \Rightarrow y = \dfrac{x^2}{3} + \dfrac{C}{x}$.

Second-Order ODEs — Complementary Function Cases

For $\dfrac{d^2y}{dx^2} + b\dfrac{dy}{dx} + cy = 0$, the auxiliary equation is $m^2 + bm + c = 0$.

Roots	CF	Behaviour

Real distinct: $m = \alpha, \beta$	$Ae^{\alpha x} + Be^{\beta x}$	Overdamped
Real equal: $m = \alpha$ (repeated)	$(A + Bx)e^{\alpha x}$	Critically damped
Complex: $m = p \pm qi$	$e^{px}(A\cos qx + B\sin qx)$	Underdamped / oscillatory
Pure imaginary: $m = \pm qi$	$A\cos qx + B\sin qx$	Simple harmonic / undamped

Example 1 (real distinct): $\dfrac{d^2y}{dx^2} - 5\dfrac{dy}{dx} + 6y = 0.$
AE: $m^2 - 5m + 6 = 0 \Rightarrow (m-2)(m-3) = 0 \Rightarrow m = 2, 3.$
CF: $y = Ae^{2x} + Be^{3x}.$

Example 2 (repeated root): $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = 0.$
AE: $(m-2)^2 = 0 \Rightarrow m = 2$ (repeated).
CF: $y = (A + Bx)e^{2x}.$

Example 3 (complex roots): $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} + 5y = 0.$
AE: $m^2 + 2m + 5 = 0 \Rightarrow m = \dfrac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i.$
CF: $y = e^{-x}(A\cos 2x + B\sin 2x).$

Particular Integrals — Worked Examples

Example 1 — Polynomial RHS: $\dfrac{d^2y}{dx^2} + 3\dfrac{dy}{dx} + 2y = 4x + 1.$
Try PI: $y_p = ax + b.\ y_p'' = 0,\ y_p' = a.$
Sub: $0 + 3a + 2(ax + b) = 4x + 1 \Rightarrow 2ax + (3a + 2b) = 4x + 1.$
$2a = 4 \Rightarrow a = 2.\ 3(2) + 2b = 1 \Rightarrow b = -\frac{5}{2}.$ PI: $y_p = 2x - \frac{5}{2}.$

Example 2 — Exponential RHS: $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = e^{4x}$.
Try $y_p = \lambda e^{4x}$. Sub: $16\lambda - 12\lambda + 2\lambda = 1 \Rightarrow 6\lambda = 1 \Rightarrow \lambda = \frac{1}{6}$. PI: $y_p = \frac{1}{6}e^{4x}$.

Example 3 — Trigonometric RHS: $\dfrac{d^2y}{dx^2} + 4y = \sin 3x$.
Try $y_p = A\cos 3x + B\sin 3x$. $y_p'' = -9A\cos 3x - 9B\sin 3x$.
Sub: $-9A\cos 3x - 9B\sin 3x + 4A\cos 3x + 4B\sin 3x = \sin 3x$.
$-5A = 0 \Rightarrow A = 0$. $-5B = 1 \Rightarrow B = -\frac{1}{5}$. PI: $y_p = -\frac{1}{5}\sin 3x$.

Boundary Conditions and Initial Value Problems

Full Example: Solve $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 3y = 6$ with $y(0) = 2,\ y'(0) = 3$.
AE: $m^2 - 4m + 3 = 0 \Rightarrow (m-1)(m-3) = 0$. CF: $y_c = Ae^x + Be^{3x}$.
PI: Try $y_p = k$ (constant). $0 - 0 + 3k = 6 \Rightarrow k = 2$. PI: $y_p = 2$.
GS: $y = Ae^x + Be^{3x} + 2$.
$y(0) = A + B + 2 = 2 \Rightarrow A + B = 0$.
$y' = Ae^x + 3Be^{3x}$. $y'(0) = A + 3B = 3$.
Solving: $B = 3/2,\ A = -3/2$.
$y = \dfrac{3}{2}e^{3x} - \dfrac{3}{2}e^x + 2$

Substitution Methods for ODEs

Some ODEs are not standard form; a substitution reduces them.

Bernoulli's equation: $\dfrac{dy}{dx} + P(x)y = Q(x)y^n$.
Substitute $v = y^{1-n}$. Then $\dfrac{dv}{dx} = (1-n)y^{-n}\dfrac{dy}{dx}$, giving a linear equation in v.

Example: $\frac{dy}{dx} - y = y^2$. Here $n = 2$. Let $v = y^{-1}$.
$\frac{dv}{dx} = -y^{-2}\frac{dy}{dx}$. Divide original by y^2: $y^{-2}\frac{dy}{dx} - y^{-1} = 1$.
So $-\frac{dv}{dx} - v = 1 \Rightarrow \frac{dv}{dx} + v = -1$. Integrating factor e^x:
$\frac{d}{dx}(ve^x) = -e^x \Rightarrow ve^x = -e^x + C \Rightarrow v = -1 + Ce^{-x}$.
$y = \frac{1}{v} = \frac{1}{Ce^{-x} - 1}$.

AQA Differential Equations — Exam Technique

- **Always write GS = CF + PI** clearly. Examiners look for this structure.
- **Resonance:** If the PI trial form already appears in the CF, multiply by x. If it appears as a double root, multiply by x^2.
- **First-order linear:** Always multiply through by IF before integrating; do not try to separate unless it genuinely is separable.
- **CF check:** Verify your CF satisfies the homogeneous equation by substituting back.
- **PI check:** Verify your PI satisfies the full equation by substituting back.
- **IVP:** Apply both initial conditions to GS to get two equations for A and B; solve simultaneously.
- **Coupled ODEs:** Eliminate one variable by differentiating one equation and substituting the other; this gives a second-order ODE in one variable.

Modelling with Differential Equations

Common models:

- **Exponential growth/decay:** $\frac{dN}{dt} = kN$ gives $N = N_0 e^{kt}$. If $k > 0$: growth; $k < 0$: decay.
- **Newton's law of cooling:** $\frac{d\theta}{dt} = -k(\theta - \theta_{\text{amb}})$. Solution: $\theta - \theta_{\text{amb}} = Ae^{-kt}$.

- **Logistic growth:** $\dfrac{dN}{dt} = kN\left(1 - \dfrac{N}{L}\right)$ where L is the carrying capacity.
- **Harmonic oscillator:** $\dfrac{d^2x}{dt^2} + \omega^2 x = 0$ gives $x = A\cos\omega t + B\sin\omega t$.
- **Damped oscillator:** $\dfrac{d^2x}{dt^2} + 2k\dfrac{dx}{dt} + \omega^2 x = 0$.

Example — Newton's cooling: A body at $80^\circ C$ cools in a room at $20^\circ C$. After 10 minutes it is $60^\circ C$. Find its temperature after 20 minutes.
$\theta - 20 = Ae^{-kt}$. At $t = 0$: $A = 60$. At $t = 10$: $40 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{2}{3}$.
At $t = 20$: $\theta - 20 = 60 \cdot \left(\frac{2}{3}\right)^2 = \frac{160}{3}$. $\theta = 20 + \frac{160}{3} \approx 73.3^\circ C$.

Resonance — Full Worked Example

Example: Solve $\dfrac{d^2y}{dx^2} + 4y = \sin 2x$ (resonance case).
CF: AE $m^2 + 4 = 0 \Rightarrow m = \pm 2i$. CF: $y_c = A\cos 2x + B\sin 2x$.
PI: Since $\sin 2x$ is in CF, try $y_p = x(C\cos 2x + D\sin 2x)$.
$y_p' = C\cos 2x + D\sin 2x + x(-2C\sin 2x + 2D\cos 2x)$
$y_p'' = -2C\sin 2x + 2D\cos 2x + (-2C\sin 2x + 2D\cos 2x) + x(-4C\cos 2x - 4D\sin 2x)$
$= -4C\sin 2x + 4D\cos 2x - 4x(C\cos 2x + D\sin 2x)$
Sub into $y_p'' + 4y_p$: $-4C\sin 2x + 4D\cos 2x = \sin 2x$.
$4D = 0 \Rightarrow D = 0$. $-4C = 1 \Rightarrow C = -\frac{1}{4}$.
PI: $y_p = -\frac{x}{4}\cos 2x$. GS: $y = A\cos 2x + B\sin 2x - \dfrac{x}{4}\cos 2x$.

Coupled ODEs — Full Worked Example

Solve: $\dfrac{dx}{dt} = x + 2y, \quad \dfrac{dy}{dt} = 3x + 2y$ with $x(0) = 1,\ y(0) = 0$.
Differentiate first equation: $\ddot{x} = \dot{x} + 2\dot{y} = \dot{x} + 2(3x + 2y) = \dot{x} + 6x + 4y$.
From first equation: $2y = \dot{x} - x \Rightarrow y = \frac{1}{2}(\dot{x} - x)$.
Sub: $\ddot{x} = \dot{x} + 6x + 2(\dot{x} - x) = 3\dot{x} + 4x$.

$\ddot{x} - 3\dot{x} - 4x = 0$. AE: $m^2 - 3m - 4 = 0 \Rightarrow (m-4)(m+1) = 0$. $m = 4, -1$.
$x = Ae^{4t} + Be^{-t}$. Then $y = \frac{1}{2}(\dot{x} - x) = \frac{1}{2}(4Ae^{4t} - Be^{-t} - Ae^{4t} - Be^{-t}) = \frac{3A}{2}e^{4t} - Be^{-t}$
.
Conditions: $x(0) = A + B = 1$; $y(0) = \frac{3A}{2} - B = 0 \Rightarrow B = \frac{3A}{2}$.
$A + \frac{3A}{2} = 1 \Rightarrow \frac{5A}{2} = 1 \Rightarrow A = \frac{2}{5},\ B = \frac{3}{5}$.
$x = \frac{2}{5}e^{4t} + \frac{3}{5}e^{-t}, \quad y = \frac{3}{5}e^{4t} - \frac{3}{5}e^{-t}$.

Second-Order ODE Applications — SHM and Damping

A mass-spring system with damping satisfies:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

where m = mass, c = damping coefficient, k = spring constant, $F(t)$ = external force.

- **Underdamped:** $c^2 < 4mk$ — complex roots, oscillatory decay.
- **Critically damped:** $c^2 = 4mk$ — repeated real roots, fastest return to equilibrium.
- **Overdamped:** $c^2 > 4mk$ — real distinct roots, exponential decay, no oscillations.

Example (underdamped): $\ddot{x} + 2\dot{x} + 5x = 0$, $x(0) = 1,\ \dot{x}(0) = 0$.
AE: $m^2 + 2m + 5 = 0 \Rightarrow m = -1 \pm 2i$. CF: $x = e^{-t}(A\cos 2t + B\sin 2t)$.
$x(0) = A = 1$. $\dot{x} = -e^{-t}(A\cos 2t + B\sin 2t) + e^{-t}(-2A\sin 2t + 2B\cos 2t)$.
$\dot{x}(0) = -A + 2B = 0 \Rightarrow B = \frac{1}{2}$.
$x = e^{-t}\left(\cos 2t + \frac{1}{2}\sin 2t\right)$.

Key Formulae Summary — Differential Equations

Equation type	Method	Key step
Separable: $\dfrac{dy}{dx} = f(x)g(y)$	Separate variables	$\displaystyle\int \frac{1}{g}\,dy = \int f\,dx$

Linear 1st order: $y' + Py = Q$	Integrating factor	$\mu = e^{\int P\,dx}$
2nd order homogeneous	Auxiliary equation	Solve $m^2 + bm + c = 0$
2nd order inhomogeneous	CF + PI	Match PI form to RHS
Coupled 1st order	Elimination	Differentiate one, sub other

Summary of PI forms:

RHS	PI trial form	If resonance
$ke^{\alpha x}$	$\lambda e^{\alpha x}$	$\lambda x e^{\alpha x}$
$k\cos\omega x$ or $k\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$	Multiply by x
kx^n	$\lambda_n x^n + \cdots + \lambda_1 x + \lambda_0$	N/A (unless $c = 0$)
$ke^{\alpha x}\cos\omega x$	$e^{\alpha x}(\lambda\cos\omega x + \mu\sin\omega x)$	Multiply by x

Euler–Cauchy Equations

The Euler–Cauchy equation $x^2\dfrac{d^2y}{dx^2} + px\dfrac{dy}{dx} + qy = 0$ is solved by the substitution $x = e^t$ (i.e. $t = \ln x$), which transforms it into a constant-coefficient ODE in t.

Example: Solve $x^2y'' - 2y = 0$.
With $x = e^t$: $x^2\dfrac{d^2y}{dx^2} = \dfrac{d^2y}{dt^2} - \dfrac{dy}{dt}$.
Equation becomes: $\dfrac{d^2y}{dt^2} - \dfrac{dy}{dt} - 2y = 0$.
AE: $m^2 - m - 2 = 0 \Rightarrow (m-2)(m+1) = 0 \Rightarrow m = 2, -1$.
GS in t: $y = Ae^{2t} + Be^{-t}$. Back-substitute $t = \ln x$: $y = Ax^2 + Bx^{-1}$.

Note for AQA: Euler–Cauchy equations may appear in AQA Further Pure papers. The substitution $x = e^t$ reduces them to constant-coefficient form. Try a power-law solution $y = x^m$ directly: substituting gives the indicial equation $m(m-1) + pm + q = 0$.

Superposition Principle

For a linear ODE $Ly = f(x)$, if y_1 is a solution when $f = f_1$ and y_2 when $f = f_2$, then $y_1 + y_2$ is a solution when $f = f_1 + f_2$.
This allows complex RHS functions to be split into simpler parts, each solved separately.

Example: To solve $y'' + y = \sin x + e^x$, find PI for $\sin x$ and PI for e^x separately, then add.
PI$_1$ for $\sin x$: since $m = \pm i$ are roots of AE, there is resonance. Try $y_{p1} = x(A\cos x + B\sin x)$.
PI$_2$ for e^x: try $y_{p2} = \lambda e^x$. $\lambda e^x + \lambda e^x = e^x \Rightarrow \lambda = \frac{1}{2}$.
GS: $y = A\cos x + B\sin x + y_{p1} + \frac{1}{2}e^x$.